Computer Networks

CS - (Fall 25)

Assignment-3

Marks 50

Submission Deadline: 23rd November 2025(11:59 PM)

Instructions:

- This is an individual Assignment. Demos will start soon after submission date, missing demo will result in straight zero marks.
- All parties involved in any kind of plagiarism/cheating will be given zero marks in all the assignments.
- Assignment deadline will not be extended.
- Only handwritten attempt will be graded, i.e., printed attempts will not be graded.
- Whenever a calculation is involved, your solution should show complete steps and a final answer.
- You must follow below submission guidelines, otherwise your submission won't be accepted
  create a new directory
  change its name to the following format YOURSECTION_ROLLNUMBER_NAME. E.g. A_20I-0012_Muhammad Ahmad
  put all your scans separately for each question
  Compress the directory into a compressed .zip file
  Submit it on google classroom

Q1. IP Addressing and Subnet Aggregation (8 Marks)

A company, TechStream Ltd., has recently acquired a block of IP addresses from its ISP to organize its internal network efficiently.

The company's data center currently uses the following four contiguous network blocks:

- 185.34.52.0/26
- 185.34.52.64/26
- 185.34.52.128/25
- 185.34.53.0/24

(a) Combine these four networks into the smallest possible single aggregate block that can represent all of them.

The ISP also assigned TechStream another block for a branch office: 198.51.120.128/26.
(b) Give one valid host IP address that can be assigned within this subnet.

The ISP owns a larger block, 198.51.120.0/24, and plans to divide it into five equal-sized subnets to allocate to its regional clients.
(c) Determine the prefixes (in the form *a.b.c.d/x*) for each of the five subnets.

Q2. NAT Operation and SDN Generalized Forwarding (12 Marks)

You are a network engineer at a small software company. The company's internal network uses private addressing in the range 192.168.10.0/24.
All Internet traffic goes through a Cisco NAT router that uses the company's single public IP address 203.0.113.15.

During work hours, the following events occur:

1. Developer PC-1 (192.168.10.25) initiates an HTTP connection (TCP port 80) to the external server 172.217.160.110 (Google) using an internal source port 51500.
2. Developer PC-2 (192.168.10.30) starts a video conference session (UDP port 5060) with the external server 52.45.33.120 (Zoom) using an internal source port 62000.
3. Later, the Google server sends a response packet back to PC-1.
4. A system administrator outside the office (45.12.88.6) attempts to connect via SSH (TCP port 22) to the internal test server 192.168.10.10 for remote troubleshooting.

Tasks

a) Construct the NAT translation table after events (1) and (2). (Show the private IP/port → public IP/port mappings.)

b) Explain how the NAT router handles the incoming packet from Google in event (3).

c) Determine whether the SSH connection from the external administrator in event (4) will succeed. Explain your reasoning.

d) Based on the concept of generalized forwarding (match + action):

I. Describe how a flow table entry on a programmable switch could be used to handle events (1-3).
II. Identify the match fields (e.g., source/destination IP, port, protocol) and the actions (e.g., forward, rewrite address/port, drop) for each event.
III. Discuss how generalized forwarding could improve flexibility or scalability in NAT operations compared to traditional destination-based forwarding.

Q3. NAT Translation (10 Marks)

You are a network analyst examining an office network that uses private addressing (10.0.0.0/24) and connects to the Internet through a NAT router with the public IP 203.0.113.50.

Each internal host communicates with external servers using either TCP or UDP. The NAT router performs port-based translation and maintains a translation table for all active flows.

During one observation period, multiple packets appear from 203.0.113.50 but with different source-port ranges, indicating that the NAT router is handling concurrent flows for multiple internal hosts.

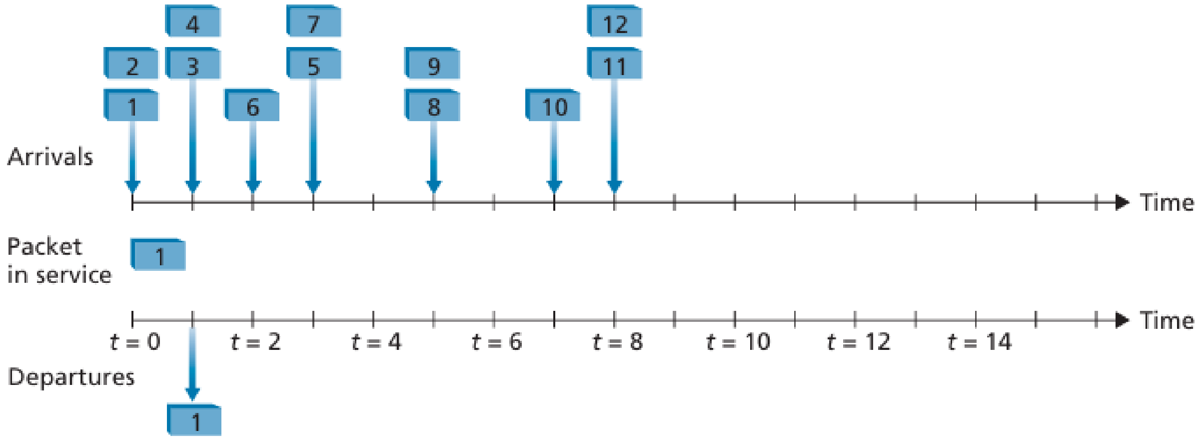
Tasks

a) Explain how the NAT device rewrites the IP and transport-layer header fields when forwarding packets from internal hosts to the Internet and when receiving the return packets.
b) Construct an example NAT translation table for three simultaneous connections from different internal hosts to different external servers (you may choose example IPs and ports).
c) If the NAT operates as a full-cone NAT, describe how an external monitoring server could send packets to internal hosts once the mappings are created.
   Using the generalized forwarding (match + action) concept:
   i. Identify the match fields and actions required to implement NAT translation for outbound and inbound traffic.
   ii. Explain how a NAT device (or router) could maintain per-flow match-action entries for active NAT mappings.

Q4. Router Queueing and Scheduling (8 Marks)

Consider the following figure showing the arrival times of packets 1 through 12 to a router. Each packet requires exactly one time slot of service. Service is non-preemptive: once a packet begins service, it runs to completion. The router is work-conserving, meaning it never stays idle when there is a packet waiting. At time t = 0, packet 1 begins service and departs at t = 1.
The remaining packets arrive at the times shown in the figure.

a) Packets 1, 4, 5, 6, and 11 belong to the high-priority class. Packets 2, 3, 7, 8, 9, 10, and 12 belong to the low-priority class. The router uses strict priority scheduling. Indicate the departure time of packets 2 through 12.
b) Now assume the router uses class-based round-robin scheduling between two classes:
   Class 1: packets 1, 4, 5, 6, 11
   Class 2: packets 2, 3, 7, 8, 9, 10, 12
   The router alternates service between the two classes. After serving one packet from Class 1, the next packet served must come from Class 2, and vice versa. If the chosen class has no waiting packets, the router serves a packet from the other class (if available). Indicate the departure time of packets 2 through 12.
c) Now assume the router implements packet-by-packet WFQ, with:
   Class 1 (packets 1, 4, 5, 6, 11): weight = 1
   Class 2 (packets 2, 3, 7, 8, 9, 10, 12): weight = 2
   Under WFQ, over time, Class 2 should receive approximately twice as many service opportunities as Class 1, while respecting packet arrival times. Indicate the departure time of packets 2 through 12.



Q5. SDN Firewall Flow Tables (12 Marks)

Consider the SDN OpenFlow topology shown below. Switch s2 should enforce firewall rules on traffic destined for hosts h3 and h4. For each of the four cases below, specify the flow-table entries in s2 that (i) permit the described traffic and forward it toward h3 or h4, and (ii) drop all other traffic destined for h3 or h4.

Do not specify any forwarding behavior for traffic destined to other hosts or routers.

a) Only traffic arriving from hosts h1 and h6 should be delivered to hosts h3 or h4
b) Only TCP traffic is allowed to be delivered to hosts h3 or h4
c) Only traffic destined to h3 is to be delivered
d) Only UDP traffic from h1 and destined to h3 is to be delivered. All other traffic is blocked